Graduate Seminar 2

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Part 1 - Solving $\psi(x,t)$

- To this end, we need to resolve $\psi(x,t)$ which is given by the regular TDSE, i.e

$$i\frac{\partial}{\partial t}\psi(x,t) = \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\psi(x,t), \qquad \psi(x,0) = \psi_0(x), \tag{1}$$

$$\hat{H}_a = -\frac{1}{2}\frac{d^2}{dx^2} - \varkappa\delta(x), \qquad \hat{V}_{\text{ext}} = F(t)x, \tag{2}$$

$$E_0 = -\varkappa^2/2, \qquad \psi_0(x) = \varkappa^{1/2}e^{-\varkappa|x|}, \qquad \varkappa = 1 \tag{3}$$

$$E_k = k^2/2, \qquad \psi_k^{(\pm)}(x) = e^{ikx} + R(\pm|k|)e^{\pm i|kx|}, \qquad R(k) = \frac{i\varkappa}{k - i\varkappa}. \tag{4}$$

- The parameters of the laser we will be using are given by the relations

$$F(t) = F_0 e^{-(2t'/\tau)^2}\cos(\omega_0 t'), \qquad 0 \leqslant t \leqslant T, \qquad t' = t - T/2, \tag{5}$$

$$\tau = 2\pi n_{oc}/\omega_0, \qquad n_{oc} = 5. \tag{6}$$

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Survival probability and PEMD

- The coefficients of the projection of $\bar{\Psi}$ are then, at T, and by definition, expressed as

$$a_0 = e^{iE_0T} \int \psi_0(x)\psi(x,T)dx, \qquad a_k = e^{iE_kT} \int \psi_k^{(-)*}(x)\psi(x,T)dx. \tag{7}$$

- The survival probability is then expressed, at the end of the interaction, by

$$P_0 = |a_0|^2 . \tag{8}$$

- The ionization probability can then be found many different ways that should be consistent with each others

$$P_{\rm ion} = \int P(k)\frac{dk}{2\pi}, \qquad \text{with} \qquad P(k) = |a_k|^2 . \tag{9}$$

$$= 1 - P_0. \tag{10}$$

- $P(k)$ is the probability for the electron being ionized with the energy E_k.

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Quantum spectrum of HHG photons (semi-classical approx.) - 1

- In the classical approx., the spectrum is obtained from the following (Eq.(80))

$$Q_\nu^{(c)}(\omega) = \alpha^2\omega^2 \left|\mathbf{A}_\nu \mathbf{d}(\omega)\right|^2, \tag{11}$$

where for an arbitrary operator $\hat{f}$, we have

$$f(t) = \int \psi^*(x,t)\hat{f}\,\psi(x,t)dx \tag{12}$$

and the Fourier transform is obtained using

$$f(\omega) = \int f(t)e^{i\omega t}dt. \tag{13}$$

- Also, we consider the relations

$$\hat{p} = -id/dx, \qquad \hat{d} = -x \tag{14}$$

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Quantum spectrum of HHG photons (semi-classical approx.) - 2

- Let us apply the following simplifications

$$\nu \longrightarrow \omega, \qquad \mathbf{A}_\nu = \sqrt{\frac{2\pi c^2}{\omega L^3}}\boldsymbol{e}_\nu \longrightarrow A_\omega, \qquad \text{and} \qquad \alpha A_\omega = 1. \tag{15}$$

- So the observable we are considering here is

$$Q^{(c)}(\omega) = \omega^2 \left|d(\omega)\right|^2. \tag{16}$$

- Additionally let us consider Eq. (104)

$$Q^{(c)}(\omega) = \left|p(\omega)\right|^2. \tag{17}$$

and, by virtue of $\boldsymbol{p}(t) = -\dot{\boldsymbol{d}}(t)$, which reads

$$Q^{(c)}(\omega) = \left|-\dot{d}(\omega)\right|^2. \tag{18}$$

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Length, velocity and acceleration gauges

- (12) applied to the operator $\hat{d}$ gives its expression in the length gauge

$$d(t) = \langle \hat{d} \rangle = -\langle \hat{x} \rangle = -\int \psi^*(x,t)\hat{x}\,\psi(x,t)dx \tag{19}$$

- $\dot{d}(t)$ is given by the formula

$$\dot{d}(t) = \frac{d}{dt}d(t) \tag{20}$$

- $p(t)$ is given by the formula

$$p(t) = \langle \hat{p} \rangle = -i\int \psi^*(x,t)\frac{\partial}{\partial x}\psi(x,t)dx \tag{21}$$

- Using the Ehrenfest theorem for the operator x, we get the relations

$$\frac{d}{dt}\langle \hat{x} \rangle = \frac{\langle \hat{p} \rangle}{m}, \tag{22}$$

$$\frac{d}{dt}\langle \hat{p} \rangle = \left\langle -\frac{\partial V_{\mathrm{ext}}}{\partial x} \right\rangle \tag{23}$$

and some others (see Bransden and Joachain p.531-532)

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Additional formula from Ehrenfest theorem

- Using the integration by parts formula, we get

$$p(\omega) = \int_{-T/2}^{T/2} e^{i\omega t} \frac{dx(t)}{dt} dt, \qquad u(t) = x(t), \quad v(t) = e^{i\omega t}, \tag{24}$$

thus

$$p(\omega) = \int_{-T/2}^{T/2} \frac{d}{dt}\Big(e^{i\omega t} x(t)\Big) dt - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt, \tag{25}$$

$$= \Big[e^{i\omega t} x(t)\Big]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt. \tag{26}$$

So, we get an additional formula for evaluating

$$Q^{(c)}(\omega) = |p(\omega)|^2\,, \tag{27}$$

with $p(\omega)$ evaluated conformly to Eq. (26).

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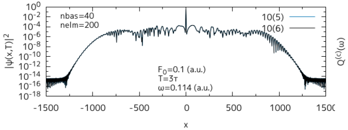
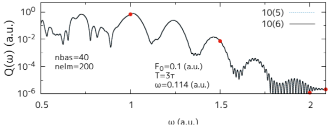
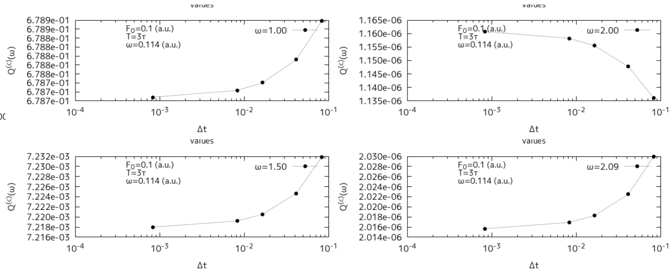
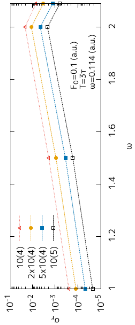
Time dependence - $T = 3\tau$







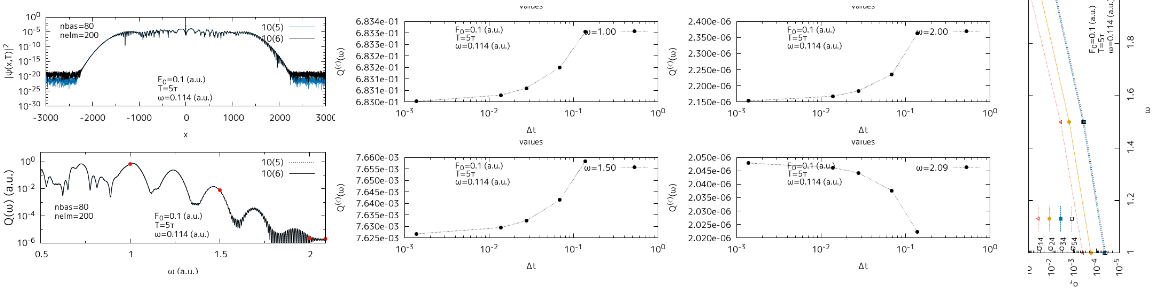


nsteps		10^4		2.10^4		5.10^4		10^5		10^6
ω	$Q^{(c)}(\omega)[1]$	σ_{15}	$Q^{(c)}(\omega)[2]$	σ_{25}	$Q^{(c)}(\omega)[3]$	σ_{35}	$Q^{(c)}(\omega)[4]$	σ_{45}	$Q^{(c)}(\omega)[5]$	σ_{55}
1.00	0.6788784	0.0002509	0.6787917	0.0001231	0.6787404	0.0000475	0.6787234	0.0000225	0.6787081	-
1.50	7.2318868(-3)	0.0019271	7.2246750(-3)	0.000928	7.2205426(-3)	0.0003554	7.2191877(-3)	0.0001677	7.2179770(-3)	-
2.00	1.1359938(-6)	0.0213401	1.1477297(-6)	0.0112297	1.1554826(-6)	0.0045506	1.1582244(-6)	0.0021885	1.1607647(-6)	-
2.09	2.0299845(-6)	0.0071059	2.0224850(-6)	0.0033853	2.0182886(-6)	0.0013034	2.0169040(-6)	0.0006165	2.0156614(-6)	-

$$\sigma_{ij} = \left| \frac{\left(Q^{(c)}(\omega)[i] - Q^{(c)}(\omega)[j]\right)}{Q^{(c)}(\omega)[j]} \right|, \qquad Q^{(c)}(\omega) = \left| \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt \right|^2 .$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Time dependence - $T = 5\tau$

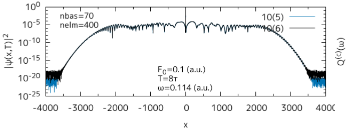


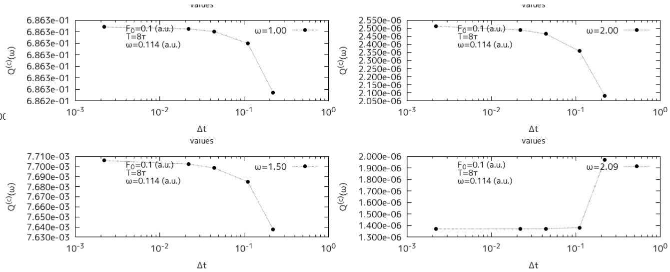
nsteps	10^4		2.10^4		5.10^4		10^5		10^6	
ω	$Q^{(c)}(\omega)[1]$	σ_{14}	$Q^{(c)}(\omega)[2]$	σ_{24}	$Q^{(c)}(\omega)[3]$	σ_{34}	$Q^{(c)}(\omega)[4]$	σ_{44}	$Q^{(c)}(\omega)[5]$	σ_{54}
1.00	0.683303	0.0004015	0.6831498	0.0001762	0.6830596	0.0000442	0.6830295	-	0.6830023	0.0000398
1.50	7.658297(-3)	0.0037774	7.6414614(-3)	0.0015707	7.6324580(-3)	0.0003906	7.6294776(-3)	-	7.6267840(-3)	-0.0003531
2.00	2.362065(-6)	0.0896783	2.2352490(-6)	0.0311749	2.1839406(-6)	0.0075051	2.1676721(-6)	-	2.1530603(-6)	-0.0067408
2.09	2.022331(-6)	-0.0710305	2.0376200(-6)	-0.0635588	2.0442287(-6)	-0.060329	2.0461706(-6)	-	2.0478578(-6)	0.0008246

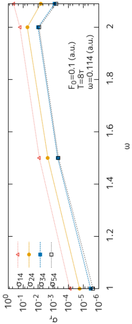
$$\sigma_{ij} = \left| \frac{\left(Q^{(c)}(\omega)[i] - Q^{(c)}(\omega)[j]\right)}{Q^{(c)}(\omega)[j]} \right|, \qquad Q^{(c)}(\omega) = \left| \Big[e^{i\omega t} x(t) \Big]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt \right|^2 .$$

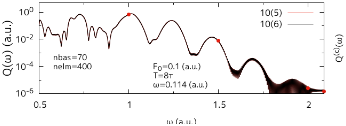
Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Time dependence - $T = 8\tau$









nsteps	10^4		2.10^4		5.10^4		10^5		10^6	
ω	$Q^{(c)}(\omega)[1]$	σ_{14}	$Q^{(c)}(\omega)[2]$	σ_{24}	$Q^{(c)}(\omega)[3]$	σ_{34}	$Q^{(c)}(\omega)[4]$	σ_{44}	$Q^{(c)}(\omega)[5]$	σ_{54}
1.00	0.6862472	-0.0000803	0.6862899	-0.0000182	0.6863001	-0.0000033	0.6863024	-	0.6863040	0.0000023
1.50	7.6376879(-3)	-0.0083983	7.6849142(-3)	-0.0022669	7.6984587(-3)	-0.0005084	7.7023749(-3)	-	7.7058210(-3)	-0.0004472
2.00	2.0810428(-6)	-0.1641447	2.3582695(-6)	-0.0527959	2.4642315(-6)	-0.0102361	2.4897164(-6)	-	2.5118399(-6)	0.0088859
2.09	1.9677690(-6)	0.4343673	1.3812389(-6)	0.0068275	1.3729975(-6)	0.0008201	1.3718724(-6)	-	1.3710046(-6)	-0.0006326

$$\sigma_{ij} = \left| \frac{\left(Q^{(c)}(\omega)[i] - Q^{(c)}(\omega)[j]\right)}{Q^{(c)}(\omega)[j]} \right|, \qquad Q^{(c)}(\omega) = \left| \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt \right|^2 .$$

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Time dependence reference table

- Sufficient time step for converged result is $\Delta t \leqslant 0.08\,\mathrm{a.u.}$
- In the following, we will maintain Δt set to around $0.04\,\mathrm{a.u.}$

$\Delta t = 0.04\,\mathrm{a.u.}$ $\omega = 0.057\,\mathrm{a.u.}$ $\tau = 551.15661$

$n = 3$		$n = 5$		$n = 8$		$n = 12$		$n = 16$	
$T = 3\tau$	nsteps	$T = 5\tau$	nsteps	$T = 8\tau$	nsteps	$T = 12\tau$	nsteps	$T = 16\tau$	nsteps
1653.4698	41336.745	2755.783	68894.576	4409.2528	110231.32	6613.8793	165346.98	8818.5057	220462.64

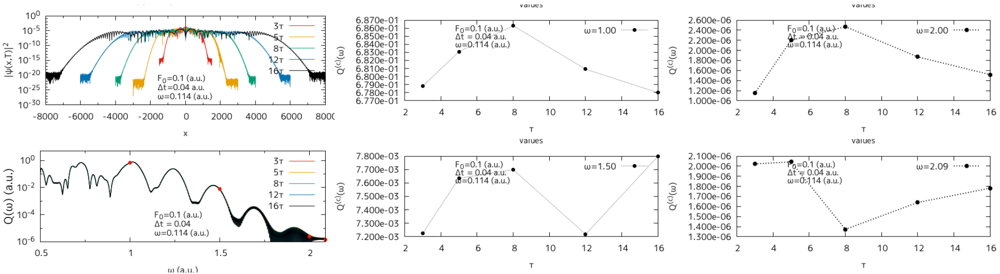
$\Delta t = 0.04\,\mathrm{a.u.}$ $\omega = 0.114\,\mathrm{a.u.}$ $\tau = 275.5783$

$n = 3$		$n = 5$		$n = 8$		$n = 12$		$n = 16$	
$T = 3\tau$	nsteps	$T = 5\tau$	nsteps	$T = 8\tau$	nsteps	$T = 12\tau$	nsteps	$T = 16\tau$	nsteps
826.73491	20668.373	1377.8915	34447.288	2204.6264	55115.661	3306.9396	82673.491	4409.2528	110231.32

$$\tau = n_{oc}\frac{2\pi}{\omega}, \qquad T = n\tau, \qquad \Delta t = \frac{T}{\text{nsteps}}, \qquad n_{oc} = 5,$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Time dependence - $F_0 = 0.1\,\mathrm{a.u.}$ $\Delta t = 0.04\,\mathrm{a.u.}$



$F_0 = 0.1\,\mathrm{a.u.}$ $\Delta t = 0.04\,\mathrm{a.u.}$, $\tau = 5 \times 2\pi/\omega = 275.5783\,\mathrm{a.u.}$						
n	$T = n\tau$	nsteps	$Q^{(c)}(\omega = 1.00)$	$Q^{(c)}(\omega = 1.50)$	$Q^{(c)}(\omega = 2.00)$	$Q^{(c)}(\omega = 2.09)$
3	826.73491	20668	0.678788979588210	7.224449588416915(-3)	1.148130921051377(-6)	2.022256187878468(-6)
5	1377.8915	34448	0.683086925522419	7.635146846170220(-3)	2.198806116374674(-6)	2.042395189629913(-6)
8	2204.288	55116	0.686300611013558	7.699195832628777(-3)	2.469101583853556(-6)	1.372771210289770(-6)
12	3306.9396	82674	0.680910244964746	7.217558844953091(-3)	1.868727433549037(-6)	1.639897289171694(-6)
16	4409.2528	110232	0.677994169556346	7.799224872958270(-3)	1.517317908696027(-6)	1.781572497404902(-6)

$$Q^{(c)}(\omega) = \left| \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t)\, dt \right|^2 .$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Time dependence - $F_0 = 0.3\,\mathrm{a.u.}$ $\Delta t = 0.04\,\mathrm{a.u.}$

$F_0 = 0.3\,\mathrm{a.u.}$ $\Delta t = 0.04\,\mathrm{a.u.}$, $\tau = 5 \times 2\pi/\omega = 275.5783\,\mathrm{a.u.}$						
n	$T = n\tau$	nsteps	$Q^{(c)}(\omega = 1.00)$	$Q^{(c)}(\omega = 1.50)$	$Q^{(c)}(\omega = 2.00)$	$Q^{(c)}(\omega = 2.09)$
3	826.73491	20668				
5	1377.8915	34448				
8	2204.288	55116				
12	3306.9396	82674				
16	4409.2528	110232				

$$Q^{(c)}(\omega) = \left| \left[e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt \right|^2 .$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Convergence parameters table for the homogeneous equation

- The following table shows parameters known to return converged results for Eq. (1), which is recalled below

$\omega = 0.057\,\mathrm{a.u.}$,	$\Delta t \leqslant 0.08\,\mathrm{a.u.}$,		$\tau = 5 \times 2\pi/\omega = 551.15661\,\mathrm{a.u.}$		
$F_0\,(\mathrm{a.u.})$	3τ	5τ	8τ	12τ	16τ
0.03	(20/200) - 3000	(60/200) - 7000	(80/200) - 10000	(90/200) - 14000*	(90-300) - 16000*
0.1	(70/400) - 8000	(100/500) - 14000	?		
0.3	?				
1.0					

Legend
$(\mathrm{nbas}/\mathrm{nelm}) - \mathrm{xmax}$

$\omega = 0.114\,\mathrm{a.u.}$,	$\Delta t \leqslant 0.08\,\mathrm{a.u.}$,		$\tau = 5 \times 2\pi/\omega = 275.5783\,\mathrm{a.u.}$		
$F_0\,(\mathrm{a.u.})$	3τ	5τ	8τ	12τ	16τ
0.03	(20/100) - 2000	(40/200) - 4000	(40/200) - 6000	(80/200) - 8000	(70-200) - 10000
0.1	(40/200) - 3000	(80/200) - 6000	(70/300) - 8000	(90/400) - 12000	(70-400) - 12000*
0.3	(90/400) - 7000	(100/500) - 10000			
1.0					

$$i\frac{\partial}{\partial t}\psi(x,t) = \left(\hat{H}_a + \hat{V}_{\mathrm{ext}}(t)\right)\psi(x,t), \qquad \psi(x,0) = \psi_0(x),$$

$$\hat{H}_a = -\frac{1}{2}\frac{d^2}{dx^2} - \varkappa\delta(x), \qquad \hat{V}_{\mathrm{ext}} = F(t)x,$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Structure of the code, performance and memory management

1. Diagonalization
   - most time consuming part for bigger matrixes
   - is there a way to multithread lapack diagonalization?
2. Propagation
   - can be arbitrarily long (convergence requires $\Delta t \leqslant 0.08\,\mathrm{a.u.}$)
   - order 4 scheme required
   - multithreading would also help here
3. Projection
   - very short (matter of minutes)
   - potentially very memory consuming

1. and 2. take some memory for a long time, 3. takes even more memory for a very short time.
Any run for large matrixes required to split those processes.

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Numerical toolkit - 1

- Integration is done by using the following order 2 methods
  - the (1/3) Simpson rule, m being the number of subintervals

$$\int_{x_0}^{x_m} f(x)dx \sim \frac{1}{3}h\sum_{i=1}^{m/2}\left[f(x_{2i-2}) + 4f(x_{2i-1}) + f(x_{2i})\right] \quad \text{(1/3 rule)} \tag{28}$$

  - the trapezoidal rule

$$\int_{x_0}^{x_m} f(x)dx \sim \sum_{i=1}^{m}\frac{f(x_{i-1}) + f(x_i)}{2}\Delta x_i \tag{29}$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Part 2 - Solving $\phi_\omega(x,t)$

- We are now interested in solving the following 1D problem with a ZRP

$$\begin{cases} i\frac{\partial}{\partial t}\psi(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\psi(x,t), \\ i\frac{\partial}{\partial t}\phi_\omega(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\phi_\omega(x,t) + e^{i\omega t}\hat{p}\,\psi(x,t), \end{cases} \tag{30}$$

where

$$\hat{H}_a = -\frac{1}{2}\frac{d^2}{dx^2} - \varkappa\delta(x), \qquad \hat{V}_{\text{ext}}(t) = F(t)x, \tag{31}$$

$$\phi_\omega(x,0) = \frac{-i\varkappa^{3/2}}{\omega}\text{sgn}(x)\left(e^{-\varkappa|x|} - e^{-\varkappa(\omega)|x|}\right), \qquad \psi(x,0) = \varkappa^{1/2}e^{-\varkappa|x|}, \tag{32}$$

with

$$\hat{p} = -id/dx, \qquad \text{and} \qquad \varkappa(\omega) = \sqrt{\varkappa^2 + 2\omega}. \tag{33}$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Solutions to the homogeneous and inhomogeneous equations

$$\begin{cases} i\frac{\partial}{\partial t}\psi(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\psi(x,t), \\ i\frac{\partial}{\partial t}\phi_\omega(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\phi_\omega(x,t) + e^{i\omega t}\hat{p}\,\psi(x,t), \end{cases}$$

$$\psi(x,t) = e^{-i\int_{t_0}^{t}(\hat{H}_a+\hat{V}_{\text{ext}}(\tau))d\tau}\psi(x,0), \tag{34}$$

$$\phi_\omega(x,t) = e^{-i\int_{t_0}^{t}(\hat{H}_a+\hat{V}_{\text{ext}}(\tau))d\tau}\phi_\omega(x,0) - ie^{-i\int_{t_0}^{t}(\hat{H}_a+\hat{V}_{\text{ext}}(\tau))d\tau}\int_{t_0}^{t} e^{i\int_{t_0}^{s}(\hat{H}_a+\hat{V}_{\text{ext}}(\tau))d\tau}e^{i\omega s}\hat{p}\,\psi(x,s)ds \tag{35}$$

$$\psi(x,t) = U(t,t_0)\psi(x,0), \qquad U(t,t_0) = e^{-i\int_{t_0}^{t}(\hat{H}_a+\hat{V}_{\text{ext}}(\tau))d\tau}$$

$$\phi_\omega(x,t) = U(t,t_0)\phi_\omega(x,0) - U(t,t_0)\int_{t_0}^{t} U(s,t_0)^{-1}e^{i\omega s}\,\frac{\partial}{\partial x}\,\psi(x,s)ds \tag{36}$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Time propagation

$$\begin{cases} i\frac{\partial}{\partial t}\psi(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\psi(x,t), \\ i\frac{\partial}{\partial t}\phi_\omega(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\phi_\omega(x,t) + e^{i\omega t}\hat{p}\,\psi(x,t), \end{cases}$$

$$\psi(x,t+\Delta t) = U(t+\Delta t,t)\psi(x,t),$$

$$\phi_\omega(x,t+\Delta t) = U(t+\Delta t,t)\phi_\omega(x,t) - \int_t^{t+\Delta t} U(t+\Delta t,s)e^{i\omega s}\,\frac{\partial}{\partial x}\,\psi(x,s)ds$$

- From the trapezoidal rule applied to the integral, we get

$$\psi(x,t+\Delta t) = U(\Delta t)\psi(x,t), \qquad U(t+\Delta t,t) = U(\Delta t) \tag{37}$$

$$\phi_\omega(x,t+\Delta t) = U(\Delta t)\phi_\omega(x,t) - \frac{\Delta t}{2}\left(e^{i\omega(t+\Delta t)}\,\frac{\partial}{\partial x}\,\psi(x,t+\Delta t) + U(\Delta t)e^{i\omega t}\,\frac{\partial}{\partial x}\,\psi(x,t)\right) \tag{38}$$

$$= U(\Delta t)\left(\phi_\omega(x,t) - \frac{\Delta t}{2}e^{i\omega t}\,\frac{\partial}{\partial x}\,\psi(x,t)\right) - \frac{\Delta t}{2}e^{i\omega(t+\Delta t)}\,\frac{\partial}{\partial x}\,\psi(x,t+\Delta t) \tag{39}$$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Projections

$$p_{k0} = \int \psi_k^{(-)*}(x)\,\hat{p}\,\psi_0(x)dx = \frac{2k\varkappa^{3/2}}{\varkappa^2 + k^2} \tag{40}$$

$$p_{kk'} = \int \psi_k^{(-)*}(x)\,\hat{p}\,\psi_{k'}^{(-)}(x)dx = 2\pi k\delta(k-k') - \frac{2\varkappa}{k^2 - k'^2 + i0}\left(\frac{k'|k|}{|k| - i\varkappa} - \frac{k|k'|}{|k'| + i\varkappa}\right) \tag{41}$$

$$b_{0\omega}(T) = e^{iE_0T}\int \psi_0(x)\phi_\omega(x,T)dx, \tag{42}$$

$$b_{k\omega}(T) = e^{iE_kT}\int \psi_k^{(-)*}(x)\phi_\omega(x,T)dx, \tag{43}$$

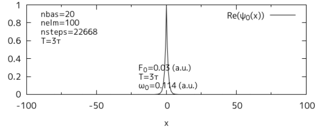
$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0+\omega-E_{k'})T}p_{0k'}a_{k'}}{E_0 + \omega - E_{k'} + i0}\frac{dk'}{2\pi} \tag{44}$$

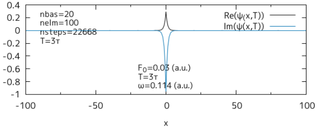
$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k+\omega-E_0)T}p_{k0}a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k+\omega-E_{k'})T}p_{kk'}a_{k'}}{E_k + \omega - E_{k'} + i0}\frac{dk'}{2\pi} \tag{45}$$

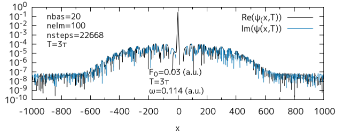
$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2\frac{dk}{2\pi} \tag{46}$$

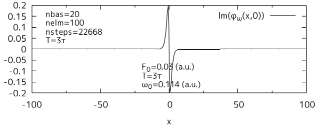
Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

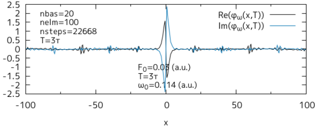
The wavefunction $\phi_\omega(x,t) \quad (\omega = 1)$

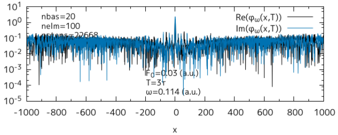


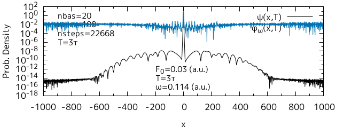












nbas $= 20$

nelm $= 100$

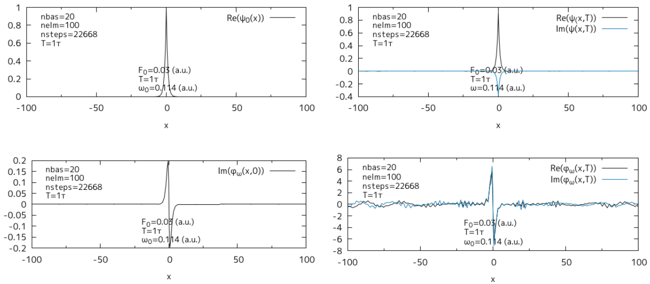
nsteps $= 22668$

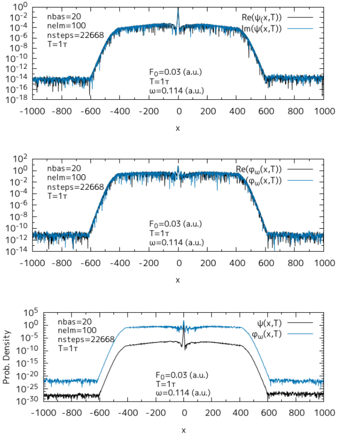
xmax $= 2000$

$T = 3\tau$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

The wavefunction $\phi_\omega(x,t) \quad (\omega = 1)$





$\mathrm{nbas} = 20$

$\mathrm{nelm} = 100$

$\mathrm{nsteps} = 22668$

$\mathrm{xmax} = 2000$

$T = 1\tau$

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

About the field